

A Beam-Conditioned 3D U-Net for Multi-Modality Radiotherapy Dose Prediction, with a Physics-Based Extension to Proton Therapy: System Design and Results for a Public Radiotherapy Dose-Prediction Grand Challenge

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Abstract—Monte Carlo (MC) dose calculation is the accuracy gold standard in radiotherapy treatment planning but is too slow for interactive or real-time use, motivating learned surrogate models that approximate MC dose distributions directly from patient imaging and beam geometry. We describe our entry to a public radiotherapy dose-prediction grand challenge across two completed tasks – Task 1 (photon dose prediction on CT) and Task 2 (photon dose prediction on MRI) against Geant4 MC ground truth – and a third, in-progress extension to proton pencil-beam-scanning therapy (Task 3). Our system combines a ray-traced photon beam encoder, which converts gantry angle, isocenter, and multi-leaf collimator (MLC) aperture into a 3D beam-path feature map, with a 3D U-Net conditioned on that map to predict per-control-point dose directly on the patient imaging grid. We train with a composite loss aligned with the challenge’s official Masked MAE metric, augmented with two auxiliary full-field terms; we present a case study in metric-driven debugging in which a systematic leakage of predicted dose mass outside the patient body – invisible to our primary training metric but catastrophic to two of the challenge’s other four scored metrics – was diagnosed, verified empirically against ground-truth data, and corrected. Task 1 and Task 2 models are trained, evaluated on all four locally computable official metrics, and packaged into Grand-Challenge-compliant inference containers, each independently validated end-to-end against a real patient input. For Task 3, we extend the framework to modality-agnostic proton pencil-beam physics – an analytic single-Gaussian beamlet model derived from the Bragg-Kleeman range-energy relation, Bortfeld range straggling, and the Highland multiple-Coulomb-scattering formula – report a GPU memory architecture fix that was necessary to make training at proton’s substantially larger voxel grid feasible at all, and present training-loss trends from an in-progress run without claiming final validated results. We report all four tasks’ status honestly, including open risks (an MR-container inference-latency measurement that has not yet been decomposed per beam, and Task 2’s early-epoch overfitting), and the concrete work remaining before final submission.

Index Terms—dose prediction, deep learning, radiotherapy, 3D U-Net, Monte Carlo surrogate, grand challenge, photon dose calculation, proton therapy, pencil beam scanning

I. INTRODUCTION

Accurate radiotherapy treatment planning requires computing the 3D radiation dose a patient will absorb for a given beam arrangement. Monte Carlo (MC) simulation, which stochastically tracks individual particle histories through a patient’s imaging-derived geometry, is the accepted accuracy standard for this calculation but is computationally expensive – a single plan can take minutes to hours to converge to low statistical noise. This cost restricts how many plan variants a clinician can practically evaluate, motivating machine-learned surrogate models that approximate MC dose output directly, at a fraction of the runtime, so that dose feedback can approach real time.

This grand challenge poses the task as a supervised learning problem across four task variants spanning two radiation modalities (photon and proton) and two imaging modalities (CT and MRI): given a patient volume and a treatment beam’s geometric parameters, predict the 3D dose distribution that a reference MC engine (Geant4) would compute, at the granularity of individual *control points* for photon therapy or individual *beamlets* for proton pencil-beam scanning. Submissions are scored on a five-metric suite spanning per-beam accuracy in the high-dose region, depth-dose-curve shape fidelity, full-plan dose accuracy stratified by dose level, a clinically-motivated 3D Gamma Index, and a DVH-based clinical score – and are further required to run within a strict per-beam inference budget on the challenge’s reference GPU hardware.

This paper documents the system we built across three of the four task variants. Tasks 1 (photon/CT) and 2 (photon/MR) are complete: both are trained, evaluated against all four locally computable official metrics on held-out patients, and packaged into tested, Grand-Challenge-compliant inference containers. We use Task 1’s first full training run as a case study in metric-

driven debugging: it produced deceptively good numbers on one metric while failing badly on others, and we walk through how we diagnosed the root cause, verified our hypothesis against real data before touching code, and corrected it – a fix that then transferred directly to Task 2 by construction, since both tasks share the same model and loss code. Task 3 (proton/CT) extends the same architecture to a physically different beam model; we describe the analytic pencil-beam physics we derived and implemented for it, a GPU memory architecture fix required to make training tractable on proton’s larger voxel grid, and report the training run’s current status without presenting unvalidated numbers as final results. Task 4 (proton/MR) has not yet been started and is discussed only as remaining work.

II. PROBLEM FORMULATION

A. Photon Tasks (1: CT, 2: MR)

Tasks 1 and 2 provide, per patient, a planning volume (CT or MRI), a set of photon beams, and for each beam a sequence of control points describing the beam’s geometry at that delivery instant (gantry angle, isocenter, and MLC leaf positions for an Elekta Versa HD linear accelerator with an Agility 160-leaf MLC). Ground-truth dose is provided per control point as a 3D volume on the same grid as the source imaging, computed by Geant4 MC simulation. The learning task is to predict this per-control-point 3D dose volume from the imaging volume and that control point’s beam parameters alone. Each (patient, beam, control point) triple is treated as an independent training example. The two tasks differ only in imaging modality and its associated preprocessing (Section III-A); the beam geometry, control-point structure, and dose grid conventions are identical, which is why a single architecture and training pipeline serves both.

B. Proton Task (3: CT)

Task 3 poses a physically different problem. Rather than control points along a continuous gantry rotation, a proton plan is described as a set of *beams*, each containing multiple *rays* (fixed source and target positions), each ray containing multiple *beamlets* at discrete energies – pencil-beam scanning (PBS) delivers dose as a superposition of many narrow, mono-energetic proton beamlets rather than a broad, MLC-shaped photon field. Ground-truth dose is again provided per beamlet as a 3D volume, but on a substantially finer, patient-specific grid ($1 \times 1 \times 3$ mm spacing, with in-plane extents up to ~ 500 voxels, versus the fixed $2 \times 2 \times 2$ mm, 256^3 -voxel grid used for Tasks 1–2). The learning task is again to predict a per-beamlet dose volume, this time from a proton beamlet’s source, target, and energy.

III. METHOD

A. Data Preprocessing

Each volume is resampled to the modality’s reference resolution (an isotropic $2 \times 2 \times 2$ mm grid for Tasks 1–2), then center cropped or padded to a fixed shape so that all examples share a common tensor shape for batching (256^3

for Tasks 1–2; $176 \times 512 \times 512$ at $3 \times 1 \times 1$ mm spacing for Task 3, matching the finer proton dose grid). CT intensities are normalized directly; MR intensities are normalized with a separate, modality-appropriate transform, since MR lacks CT’s absolute Hounsfield-unit scale. A body mask is derived directly from the source imaging in both cases (a CT-appropriate threshold for Task 1, an MR-appropriate one for Task 2), used both as an auxiliary training signal (Section III-E) and, for Task 3, as the reference surface a proton beamlet’s entry depth is measured from (Section III-C). Ground-truth dose values are scaled by a fixed constant, chosen so that typical per-control-point dose maxima (empirically $\sim 5 \times 10^{-5}$ to 1.3×10^{-4} Gy, measured directly from 120 real dose files) land near $O(1)$, so that regression targets sit in a numerically well-conditioned range for the network’s unconstrained output.

B. Photon Beam Geometry Encoding

The photon beam encoder converts a control point’s geometric parameters – gantry angle, isocenter position, and per-leaf MLC aperture boundaries – into a dense 3D feature map aligned with the imaging grid, via geometric ray tracing: for each voxel, the encoder determines whether that voxel lies within the beam’s rectangular field as delimited by the current MLC leaf positions, projected along the beam’s central axis at the given gantry angle. This produces a soft-edged beam-path mask volume that is concatenated with the source imaging as a second input channel to the network. The coordinate grid used for ray tracing depends only on volume shape and voxel spacing, both of which are fixed identically across every patient after the preprocessing in Section III-A; we therefore build this grid once and share it across all patients, applying each patient’s isocenter as a cheap per-sample offset rather than rebuilding the grid per sample. Combined with GPU execution of the ray-tracing step, this gave an approximately $78\times$ speedup over a naive per-sample CPU implementation, which was necessary to keep both training throughput and per-beam inference latency within budget.

C. Proton Beam Geometry Encoding

Task 3’s beamlet geometry – a source, a target, and a single energy – admits no MLC-aperture analogue, so we designed an analytic pencil-beam encoder from first-principles proton physics rather than reusing the photon ray-tracing approach. For a beamlet of energy E (MeV), we compute:

- **Range** via the Bragg-Kleeman relation, $R[\text{mm}] = 10 \cdot 0.0022 \cdot E[\text{MeV}]^{1.77}$;
- **Range straggling** via Bortfeld’s mono-energetic approximation [4], $\sigma_R[\text{mm}] = 10 \cdot 0.012 \cdot R[\text{cm}]^{0.935}$;
- **Lateral spread** via the Highland multiple-Coulomb-scattering formula [3], evaluated in water ($X_0 = 360.8$ mm) using relativistic beam kinematics for a proton (rest mass 938.272 MeV) at energy E .

For each voxel, we compute its signed depth along the beamlet axis and its lateral distance from that axis, then evaluate a single Gaussian in depth (centered at the entry-to-body depth plus range, width σ_R) and a single Gaussian in the lateral

direction (width given by the Highland formula), matching the single-Gaussian pencil-beam approximation used in the challenge’s own reference beam model. Entry-to-body depth is found per ray by marching the depth axis against the body mask from Section III-A, since a beamlet’s source point sits in air outside the patient rather than at the body surface – an early version of this encoder measured range from the source instead of the body entry point and produced a systematically wrong (all-zero, after range-gating) mask; we caught this by validating peak-dose-location distance against real beamlet dose files (3–17 mm accuracy across five sampled beamlets) before trusting the encoder in training.

1) *A shared-coordinate-grid memory fix, discovered by Task 3:* Both encoders were originally designed to cache one full-volume coordinate-grid tensor per *patient*, since coordinate grids are expensive enough to build that avoiding a rebuild per sample was essential for training throughput (Section III-B). This was safe for Task 1–2’s fixed 256^3 grid (~ 200 MB per cached tensor, ~ 13.4 GB across 67 train patients) but fatal for Task 3’s much larger $176 \times 512 \times 512$ grid (~ 553 MB per tensor, ~ 37 GB across the same number of patients) – Task 3 training crashed with a CUDA out-of-memory error partway through its first epoch, after enough distinct shuffled patients had been visited to fill the cache. We root-caused this via the memory arithmetic above rather than treating it as a generic “reduce batch size” problem, and fixed it by recognizing that the coordinate grid itself is patient-*independent* once volume shape and voxel spacing are fixed by preprocessing (Section III-A) – only the per-patient isocenter/origin offset actually varies, and that is a cheap (3,) tensor rather than a full volume. Refactoring both encoders to build one shared, origin-free coordinate grid and apply the per-patient origin as a lightweight offset collapsed the cache to a single shared entry, eliminating the OOM without changing model output (re-validated: identical beamlet peak-distance numbers before and after the refactor).

D. Model Architecture

We use a 3D U-Net (five-level encoder/decoder, channel widths 16–32–64–128–256, stride-2 downsampling at each level, two residual units per block) taking the two-channel (source imaging, beam mask) volume as input and predicting a single-channel dose volume of the same spatial size. Beam conditioning is early-fused: the beam mask is treated as a second input channel alongside the imaging volume rather than injected at a later layer, so that the network’s earliest convolutional features already have access to beam geometry. The identical architecture is used, unmodified, for all three tasks; only the input preprocessing and beam encoder differ.

E. Training Objective

Our loss function is designed around the challenge’s own primary per-beam metric, *Masked MAE*: mean absolute error restricted to voxels in the high-dose region (dose $\geq 10\%$ of that beam’s maximum dose), normalized by that maximum. We train a directly differentiable version of this exact quantity

rather than a loose proxy, so that the training objective and the scored metric are aligned as closely as possible.

Optimizing Masked MAE alone gives the network no gradient signal outside the high-dose region, which risked the network treating everything outside that mask as unconstrained – fine for Masked MAE itself, but damaging to metrics that examine the full dose distribution (the Gamma Index and DVH-based clinical score in particular). We therefore add an auxiliary full-field term restricted to voxels inside the patient body mask, giving the network gradient signal in the low- and mid-dose region as well.

a) *An out-of-body dose leakage bug, diagnosed by metric, not by inspection:* After Task 1’s first full training run (10 epochs), Masked MAE looked reasonable ($\approx 9.5\%$), but two of the other four official metrics – IDD curve distance and Gamma Index pass rate – were badly degraded, in a way inconsistent with a model that was simply undertrained (Table I). We hypothesized that the two loss terms above, which only ever examine voxels *inside* the high-dose mask or the body mask respectively, gave the network no reason to suppress dose predicted *outside* the patient’s body – a region that never contributes to the training loss regardless of how much (incorrect) dose mass the network places there, but does contribute to metrics that look at the full 3D distribution or an along-axis depth-dose curve. Before changing any code, we verified this hypothesis directly against real validation data: ground-truth dose outside the body mask measured $\approx 1 \times 10^{-8}$ on average across sampled control points, five to six orders of magnitude below typical in-body dose maxima ($\approx 10^{-4}$), confirming that dose outside the body is expected to be physically negligible and that any systematic leakage there is a genuine model defect rather than a property of the data.

We corrected this by adding a third loss term that penalizes the model’s mean absolute error against (near-zero) ground truth specifically in the region outside the body mask, normalized the same way as the other two terms. The combined objective is

$$\mathcal{L} = w_m \mathcal{L}_{\text{masked}} + w_b \mathcal{L}_{\text{body}} + w_o \mathcal{L}_{\text{outside}}, \quad (1)$$

with $w_m = 1.0$ and $w_b = w_o = 0.2$ – the metric-aligned term dominates, with the other two acting as smaller regularizers. Because this fix lives entirely in shared loss/training code, it applied to Task 2 automatically without any Task-2-specific change.

F. Training Procedure

We optimize with AdamW (learning rate 2×10^{-4} , weight decay 1×10^{-5}) and a cosine-annealing learning-rate schedule, using automatic mixed precision and gradient-norm clipping (max norm 1.0) for stability. Batch size is fixed at 1 by design (each example requires its own beam-encoder call regardless of batching, since encoders are not vectorized across samples with different origins), with gradient accumulation available for a larger effective batch. Patient identity, not control-point/beamlet identity, defines the train/validation split (90/10), so that validation measures generalization to unseen patients.

TABLE I
TASK 1 (PHOTON/CT) VALIDATION METRICS, PRE- VS. POST-FIX

Metric	Pre-fix	Post-fix
Masked MAE	0.0953 (9.5%)	0.0949 (9.5%)
IDD curve distance	126.5	0.56
Stratified Plan MAE	0.0716 (7.2%)	0.0509 (5.1%)
Gamma pass rate (1%/1 mm)	0.0135 (1.3%)	0.0328 (3.3%)

A finite-value check before both the backward pass and the optimizer step skips (rather than applies) any step producing non-finite loss or gradients, since an earlier unguarded run silently degraded to a constant NaN output partway through its first epoch, undetected for the remainder of a multi-day run.

G. Evaluation Protocol

We evaluate against all four of the challenge’s locally quantifiable official metrics on held-out patients: Masked MAE (per beam); IDD (Integrated Depth-Dose) curve distance, the root-mean-square deviation between predicted and ground-truth dose profiles along the beam’s central axis, normalized by peak ground-truth IDD; Stratified Plan MAE, a full-plan (all of a patient’s beam doses summed) L1 error computed separately within high/mid/low dose bands relative to a prescription-dose reference; and 3D local Gamma Index at the clinically strict 1%/1 mm criterion, using each plan’s own maximum dose as a prescription-dose stand-in. The fifth official metric, a DVH-based clinical score, requires structure/contour data (PTV and organ-at-risk delineations) not present in the locally available training data; we report it as unavailable rather than approximate it with a substitute that could misrepresent clinical relevance.

IV. RESULTS

A. Task 1: Photon/CT

Table I reports validation-set metrics (8 held-out patients, 540 control points each) before and after the out-of-body correction of Section III-E, both evaluated from a fine-tune resumed at the pre-fix run’s best checkpoint. The fix was confirmed effective *before* the full fine-tune completed, by evaluating the first post-fix epoch alone.

IDD curve distance improved roughly $225\times$ and now sits solidly in the expected $O(10^{-2}-10^0)$ range; Stratified Plan MAE and Gamma pass rate both improved substantially as well, while Masked MAE (the only metric the pre-fix model was already directly optimizing inside the high-dose mask) was essentially unchanged, exactly as expected for a fix that targets an out-of-mask failure mode. Gamma pass rate remains low in absolute terms; we believe this is at least partially an artifact of our local prescription-dose stand-in (each plan’s own maximum dose, in the absence of real RTSTRUCT prescription data locally), rather than purely a model-accuracy issue, since the Gamma Index computation is sensitive to this reference value – but we have not yet been able to confirm this against the real challenge test set’s prescription data. The

TABLE II
TASK 2 (PHOTON/MR) VALIDATION METRICS

Metric	Value
Masked MAE	0.1025 (10.3%)
IDD curve distance	4.32
Stratified Plan MAE	0.0920 (9.2%)
Gamma pass rate (1%/1 mm)	0.0309 (3.1%)

full 10-epoch fine-tune completed without further crashes; validation loss was lowest at the first post-fix epoch, with mild re-overfitting in later epochs, so the reported checkpoint is that first post-fix epoch, not the final one.

B. Task 2: Photon/MR

Table II reports Task 2 validation metrics (same 8 held-out patients, 540 control points each) using the best checkpoint by validation loss.

The best Task 2 checkpoint occurred at the first training epoch; validation loss degraded steadily in every subsequent epoch while training loss continued to improve smoothly, a clear overfitting signature we detected by comparing the two curves rather than trusting training loss alone, and which we handle correctly by construction since our checkpointing logic already always saves the best-by-validation-loss checkpoint separately from the most recent one. Gamma pass rate (3.1%) is close to Task 1’s post-fix value (3.3%), consistent with the same local prescription-approximation limitation discussed above rather than an MR-specific problem; IDD curve distance is higher than Task 1’s post-fix value but still in a qualitatively reasonable range, and Masked MAE and Stratified Plan MAE are both modestly higher than Task 1, plausibly reflecting the shorter effective training (one well-generalizing epoch, versus Task 1’s full fine-tune) rather than an inherent MR-versus-CT accuracy gap – distinguishing between these two explanations is listed as remaining work (Section VI).

C. Inference Speed

Task 1’s measured per-control-point inference latency inside the packaged submission container, after profiling and removing three avoidable costs (a redundant CPU array copy, a duplicate SimpleITK round-trip for cutoff thresholding, and unwarmed CUDA/cuDNN kernel selection on the first real call), is ≈ 0.16 s/control point on our development GPU (NVIDIA A100) – roughly $6\times$ under the challenge’s 1-second-per-beam limit. Task 2’s container was validated end-to-end against a real patient input (health check, warmup, and a successful `/invoke` call producing correct output), completing in 1.94 s total for that call; unlike Task 1, this has not yet been decomposed into a controlled per-control-point measurement, so whether Task 2 meets the same per-beam budget with the same margin is an open question we flag explicitly rather than assume by analogy to Task 1. Timing on the challenge’s actual reference hardware (an AWS g5 instance) has not yet been independently confirmed for either task.

D. Submission Packaging

Both Task 1 and Task 2 are packaged behind the challenge’s required HTTP `invoke` API (a `/health` endpoint the platform polls until the model is loaded and warmed, and a `/invoke` endpoint that executes one prediction call) inside independent Docker containers, following the challenge’s documented input/output contract: imaging and beam-metadata inputs are provided in fixed, always-fully-populated 10-slot batches (unused slots contain $1 \times 1 \times 1$ placeholder volumes), and predicted dose is written back in the same 10-slot structure, stacked into genuine 4D volumes via SimpleITK’s `JoinSeries` rather than a raw array stack (which the challenge documentation notes can silently corrupt multi-beam output), on a grid that exactly matches each input volume’s size, spacing, origin, and direction. Both containers were validated end-to-end locally against real patient input, confirming correct startup, correct output geometry and content, and (for Task 1) per-beam latency well under budget.

V. EXTENSION TO PROTON THERAPY (TASK 3)

Task 3 reuses the architecture, loss, and training procedure of Tasks 1–2 unmodified, substituting the proton pencil-beam encoder of Section III-C and the finer $176 \times 512 \times 512$ grid preprocessing of Section III-A. The full 423 GB proton training dataset (spanning ~ 77 patients and $\sim 81,000$ beamlets, matching the challenge’s documented scale) was downloaded and verified against real patient geometry (grid spacing, extent, and beam JSON schema cross-checked on five real patients) before training began. Training is ongoing at the time of writing. We report training-loss trends only, and explicitly do not present a validated held-out result for this task: the first training epoch completed with a validation loss of 0.1032, and a second epoch is in progress; given Task 2’s demonstrated overfitting pattern (best result at the very first epoch), our stopping criterion is to compare each subsequent epoch’s validation loss against the running best and stop once it degrades, rather than committing in advance to a fixed epoch count. Task 3 evaluation code (extending `evaluate.py` to the proton beamlet index/encoder, analogous to but not identical in structure to the photon evaluation path) and submission packaging have not yet been built.

VI. REMAINING WORK

The following items remain, in priority order:

- 1) **Confirm Task 2’s per-control-point inference latency** against the 1-second-per-beam budget with the same rigor applied to Task 1, rather than relying on a single aggregate `/invoke` timing.
- 2) **Upload the completed Task 1 and Task 2 containers** to the Grand Challenge platform – the platform’s actual build/run/scoring behavior has not yet been exercised end-to-end with our real account, and is currently the largest untested risk for both completed tasks.
- 3) **Complete or early-stop Task 3 training**, run its full evaluation suite, and determine whether Task 3’s (currently unconfirmed) accuracy and inference latency

meet the same bar as Tasks 1–2 before committing engineering time to its submission container.

- 4) **Build Task 3’s submission container**, which will require different input parsing (beam/ray/beamlet JSON rather than control-point JSON) than the shared Task 1–2 container pattern.
- 5) **Task 4 (proton/MR)** has not yet been started; it would reuse Task 3’s proton physics with Task 2’s MR pre-processing, by the same composition principle already demonstrated between Tasks 1 and 2.
- 6) **Investigate the Gamma Index pass-rate gap** between our local prescription-stand-in evaluation and the real challenge test set’s prescription-based scoring, to determine how much of the currently low pass rate (Tasks 1–2, 3.1–3.3%) reflects genuine model inaccuracy versus an artifact of the local evaluation approximation.

VII. CONCLUSION

We have described a beam-conditioned 3D U-Net system for radiotherapy dose prediction spanning two photon imaging modalities (CT and MRI) for a public grand challenge, and a physics-based extension to proton pencil-beam scanning. Using Task 1’s first full training run’s evaluation results as a diagnostic signal rather than a final verdict, we identified, empirically verified, and corrected a specific failure mode – dose leakage outside the patient body – via an additional loss term that then transferred to Task 2 without modification, by virtue of a shared codebase. We further described a proton pencil-beam encoder derived from established beam physics (Bragg-Kleeman range, Bortfeld straggling, Highland scattering), and a GPU memory architecture fix required to make its training tractable at proton’s substantially larger voxel grid. We report Tasks 1 and 2 as complete, evaluated, and packaged for submission, and Task 3 as an in-progress extension with training under way but no validated held-out results yet, and Task 4 as not yet started. We view both the diagnostic process behind the Task 1–2 fix and the first-principles physics derivation behind the Task 3 encoder as useful methodological contributions independent of final leaderboard standing.

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and Section IV’s narrative figures) were produced by running the author’s own training and evaluation code against the challenge’s official dataset and metrics; no results in this paper were generated or fabricated by the AI assistant.

APPENDIX

ARTIFACT DESCRIPTION (AD) APPENDIX

Computational environment. All training and evaluation was run on a single NVIDIA A100 (40 GB) GPU, with model code implemented in PyTorch. Container packaging targets the Grand Challenge platform’s documented reference hardware (an AWS g5 instance).

Software. PyTorch (mixed-precision training via `torch.amp`), SimpleITK (medical image I/O and resampling), Docker (submission packaging, following the challenge’s required `invoke` HTTP API contract).

Data. The challenge’s official training data (photon: 77 patients $\times$ 3 beams $\times$ 180 control points; proton: $\sim$ 77 patients, $\sim$ 81,000 beamlets), obtained from the challenge organizers under the challenge’s data-use terms. This data is not independently redistributable by the authors.

Availability. Source code (training pipeline, beam encoders, loss functions, evaluation harness, and Docker packaging scripts) is maintained in a private repository at submission time and has not yet been made publicly available; the authors intend to release it following the challenge’s submission deadline, subject to any challenge-specific terms governing code release. No public artifact identifier (DOI, public repository URL) is available to include in this submission for that reason.

Reproducibility notes. Exact hyperparameters (learning rate, loss term weights, batch/accumulation configuration, epoch counts per task) are given in Sections III-E–III-F and IV; the out-of-body loss fix (Section III-D) and the shared-coordinate-grid memory fix (Section III-C-1) are both described with enough detail to reimplement independently of our specific codebase.

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